

IDBI INNOVATE 2026

Prospect Assist AI

AI-driven lead intelligence for retail lending — replacing self-reported income with behavior-verified repayment capacity and conversion-ready leads.

TEAM NAME

[Your Team Name]

TEAM LEADER

[Team Leader Name]

PROBLEM STATEMENT

PS 2 — Prospect Assist AI

Build. Integrate. Transform.

Knowledge Partner: AWS | Powered by: H2S | Technology Partner: ACC

Brief about the Idea

Prospect Assist AI turns every customer's existing bank relationship into a live underwriting signal. Instead of scoring leads on declared income and demographic guesswork, it reads salary credit regularity, UPI spend patterns, savings growth, existing EMI load, and recent big-ticket behaviour (e.g. property or dealer-related transfers) to answer two questions banks actually need answered before they call a prospect:

Can they genuinely repay?

A behavior-verified income & repayment-capacity estimate — not a self-reported number.

Are they actually interested — right now?

A conversion-probability score built from real transaction intent signals, refreshed continuously.

TARGETED OUTCOME

>30%

conversion rate target set by IDBI Bank for generated leads

55.36%

achieved by our prototype model on held-out synthetic data — nearly 2x the base conversion rate

61.58%

more accurate income estimation than trusting self-declared income

Opportunities

1

How is this different?

Most lead-scoring tools re-package CRM/demographic data or trust declared income. Prospect Assist AI is behavior-first: it derives repayment capacity and intent directly from transaction cash-flow, closing the gap between what a customer says and what their account actually shows.

2

How does it solve the problem?

It replaces two weak, manual signals (declared income, generic demographics) with three model-driven ones — verified income, a calibrated conversion probability, and a product-fit recommendation — so RMs call the right customer, about the right product, with the right amount in mind.

3

USP of the solution

Explainable-by-design (SHAP-style contribution for every score, audit-ready for underwriting), tunable to the bank's own risk appetite via one threshold, and deployable directly on IDBI's sandbox APIs and synthetic datasets with no architecture change.

List of Features Offered by the Solution

1

Behavior-Verified Income

Estimates true monthly income from cash-flow patterns — independent of what's declared on the form.

2

Real-Time Lead Scoring

0–100 conversion-probability score per customer, recomputed as new transactions arrive.

3

Product-Fit Recommendation

Auto-matches each lead to Personal / Home / Auto / Mortgage Loan based on behavioral fit.

4

Explainable Scoring

Every prediction ships with top feature contributions — audit-ready for underwriting.

5

RM Lead Queue Dashboard

Ranked, filterable, exportable list of high-quality leads for relationship managers.

6

Portfolio Analytics

City-wise, product-wise lead distribution and declared-vs-estimated income gap analysis.

7

Configurable Risk Threshold

One tunable cutoff lets the bank trade off lead volume vs. conversion quality.

8

Sandbox-Ready Integration

Built to plug directly into IDBI Bank's sandbox APIs and synthetic datasets.

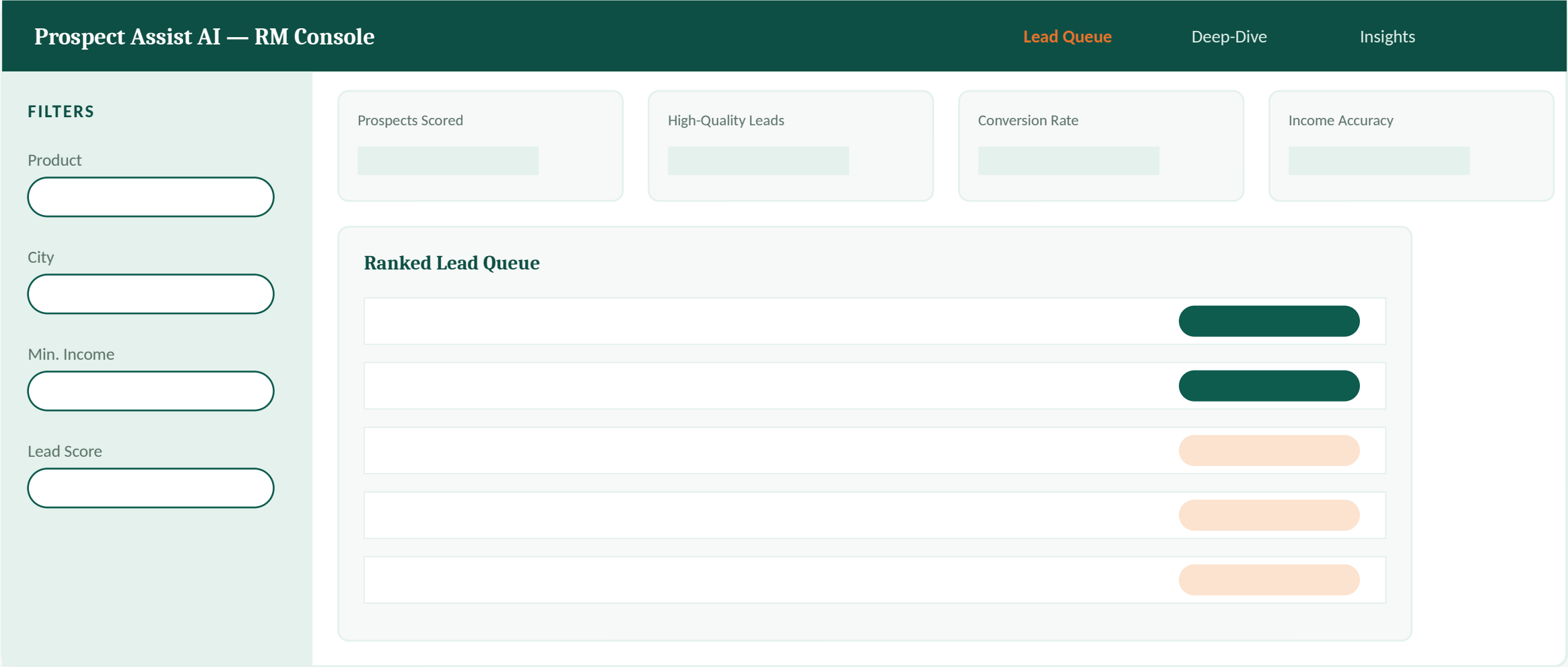
Process Flow Diagram



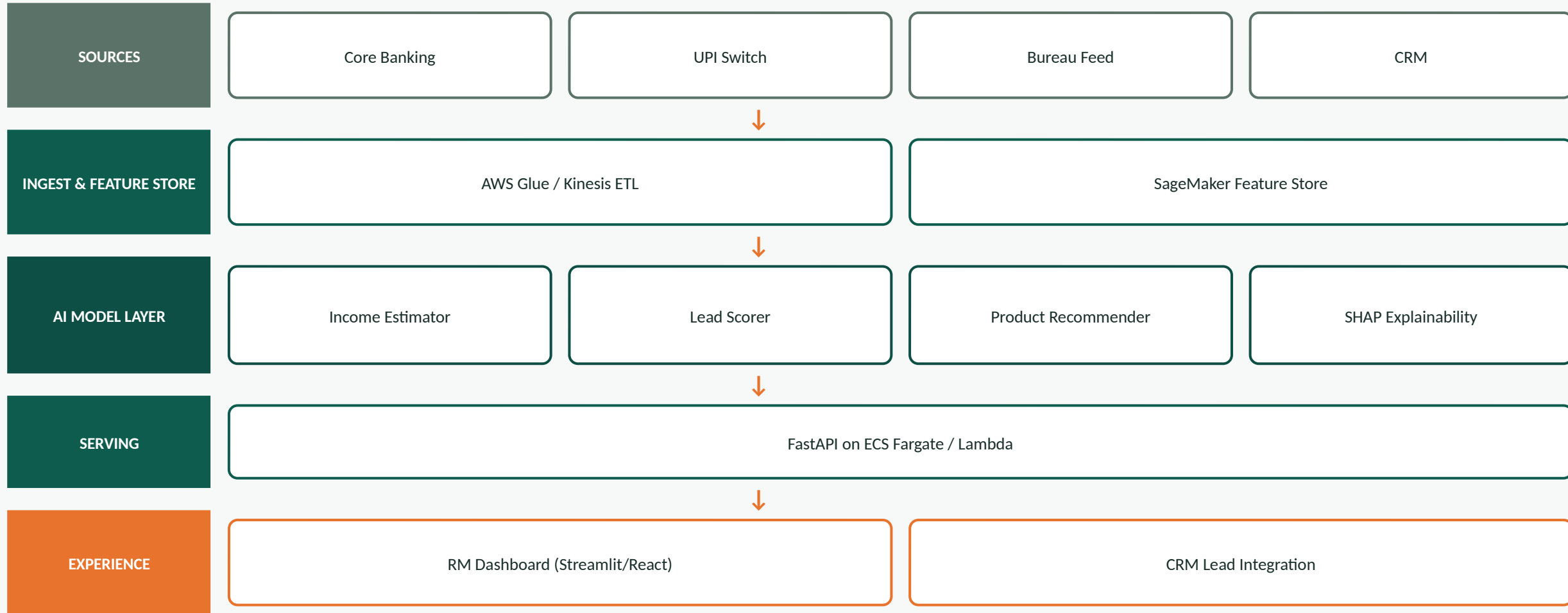
Loop: Outcome Feedback (6) continuously retrains the AI Scoring Engine (3), improving precision each cycle.

Use-case: Retail RM logs in every morning to a queue of the day's highest-probability, behavior-verified leads — instead of cold-calling a static branch list.

Wireframes of the Proposed Solution



Architecture Diagram of the Proposed Solution



Feedback loop: RM-confirmed outcomes (converted / not) flow back into SageMaker Pipelines to retrain all three models on a rolling schedule.

Technologies Used in the Solution

Data & ML

Python

Pandas / NumPy

XGBoost

SHAP

scikit-learn

Serving & App

FastAPI

Streamlit

Plotly

Joblib

Cloud & Infra (production)

AWS SageMaker

AWS Lambda / ECS Fargate

AWS Glue / Kinesis

AWS S3 / Redshift

Bank Integration

IDBI Sandbox APIs

Synthetic Banking Datasets

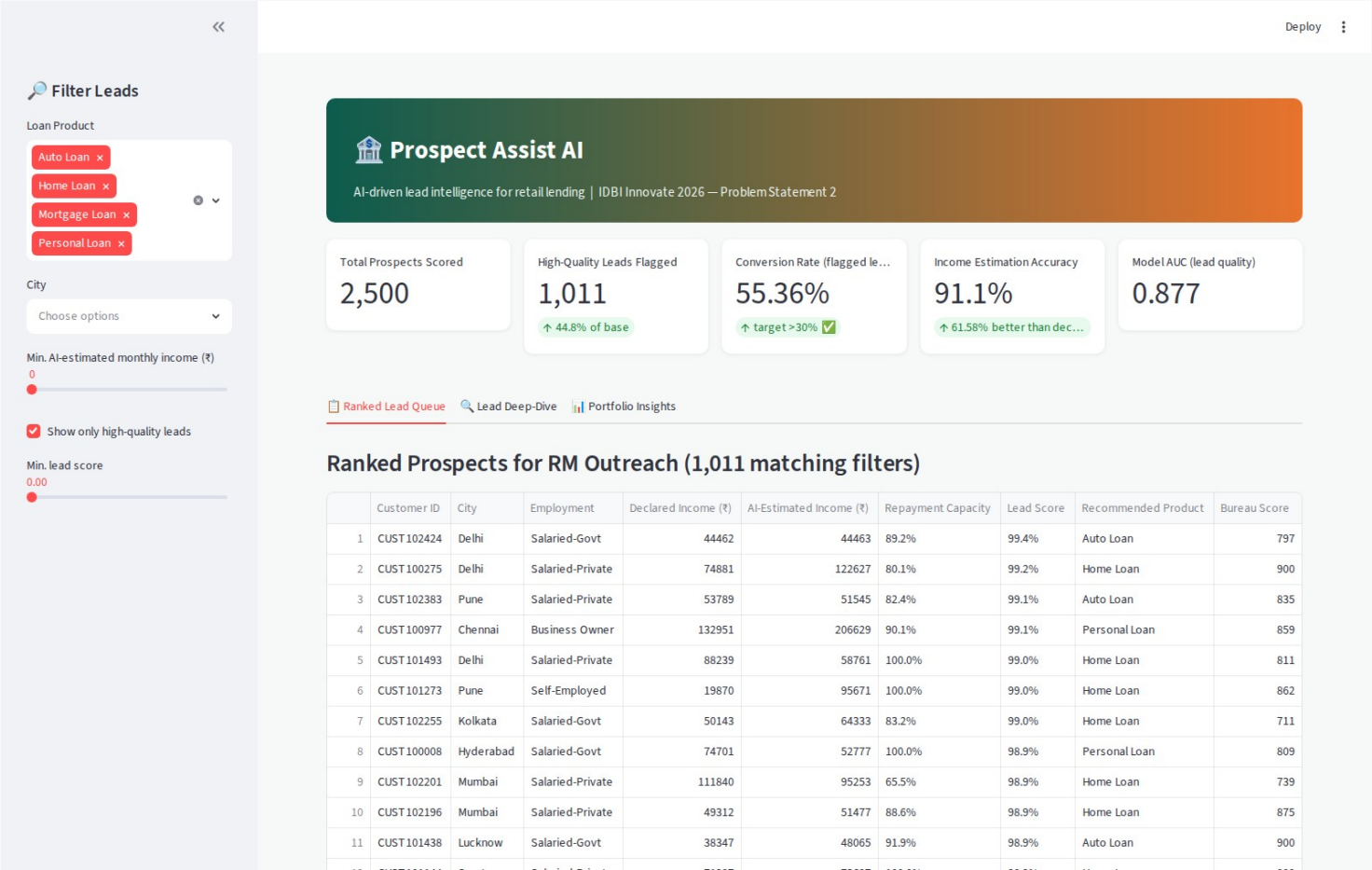
CRM / Lead Management
System

Estimated Implementation Cost (Optional)

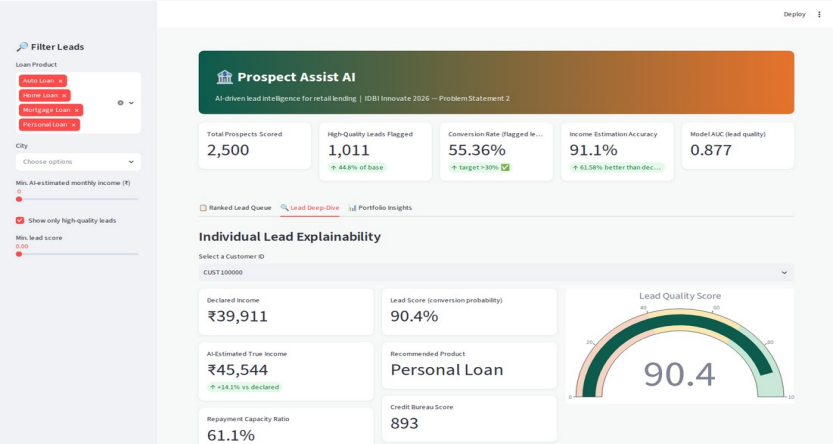
Phase	Scope	Est. Cost
Sandbox PoC (3 months)	AWS SageMaker + Lambda/Fargate (dev-tier), 2 ML engineers, 1 designer	₹9 – 12 L
Pilot at 2–3 branches (6 months)	Model retraining pipeline, CRM integration, monitoring & MLOps	₹18 – 25 L
Bank-wide rollout (annual run-rate)	Production AWS infra, compliance/audit tooling, ongoing model ops	₹35 – 50 L / yr

Indicative, hackathon-stage estimate for planning discussion only — final costing depends on IDBI Bank's sandbox infra terms and data-access scope.

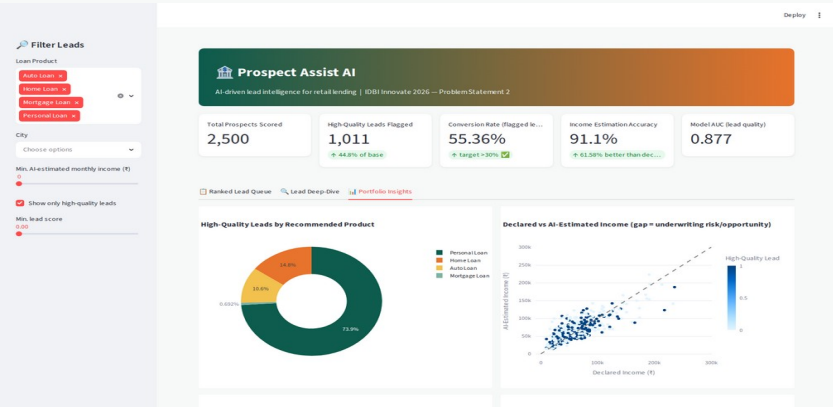
Snapshots of the Prototype



Ranked Lead Queue



Lead Deep-Dive & Explainability



Portfolio Insights

Prototype Performance Report / Benchmarking

55.36%

Conversion rate among AI-flagged leads

vs. 30% target — nearly 2x lift over base rate

0.8767

Lead Scorer AUC-ROC

on held-out synthetic test data

61.58%

More accurate than declared income

MAE ₹5,789 vs ₹15,066 for the declared-income baseline

94.5%

Income variance explained (R^2)

by the behavior-only income estimator

98.8%

Product-recommendation accuracy

Personal / Home / Auto / Mortgage classification

44.8%

Of base flagged as high-quality leads

large enough pipeline for RM teams to act on daily

Additional Details / Future Development

1

Bureau + GST data fusion

Blend bureau pulls and GST returns (for self-employed/MSME) with transaction features for even sharper income estimation.

2

Real-time scoring API

Move from batch scoring to an event-driven API that re-scores a customer the moment a qualifying transaction lands.

3

Active-learning feedback loop

RM-confirmed conversion outcomes automatically retrain the models weekly, closing the loop shown in the architecture.

4

Multi-lingual RM assistant

A conversational layer so RMs can ask "who are my top 10 home-loan leads in Pune this week" in natural language.

5

Fairness & bias monitoring

Continuous fairness audits across gender, geography, and employment type before any production rollout.

6

Early-warning cross-sell

Extend the same behavioral engine to flag credit-card, FD, and insurance cross-sell opportunities.

Provide Links To Your

1

GitHub Public Repository

[paste your public repo URL here]

2

Demo Video Link (3 Minutes)

[paste your unlisted/public video URL here]

3

Final Product Link

[paste your deployed Streamlit Cloud URL here]

Thank You

Prospect Assist AI — Team [Your Team Name]

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